

Process Characterization Report

**A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) —
PCR-008**

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, one-sided process capability index; CQA, critical quality attribute; CV, coefficient of variation; DoE, design of experiments; DNA, deoxyribonucleic acid; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LOQ, limit of quantitation; LRF, log reduction factor; mAU, milli-absorbance unit; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSM, response surface methodology; SDM, scale-down model; SEC, size exclusion chromatography; SOP, standard operating procedure; TCID₅₀, fifty percent tissue culture infective dose; UV, ultraviolet; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Anion exchange chromatography is the final chromatographic step in the A-Mab drug substance process. It is operated in flow-through mode, so the antibody passes through the column while host cell protein, DNA, leached Protein A and virus bind to the resin. This report presents the characterization of the step against its approved plan (PCP-008), executed on a qualified scale-down model at commercial equivalent conditions. The work follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), and the viral clearance evaluation follows ICH Q5A(R2) (International Council for Harmonisation 2023a). The process model, the criticality framework and the modular clearance approach are taken from the A-Mab case study (CMC Biotech Working Group 2009).

5 process parameters were carried into the characterization, 4 of them in a multivariate design and 1 univariately. All of them are classified as well controlled critical process parameters. The step has no critical process parameter and no key process parameter. The designed experiments were a screening study of 19 runs and a face-centred response-surface study of 28 runs, both executed on the same scale-down column.

The critical quality attribute assigned to this step is the clearance of minute virus of mice (MVM), which the drug substance register rates of very high criticality. The step also governs pool host cell protein and contributes to the clearance of xenotropic murine leukaemia virus (XMuLV). Response-surface models for these three responses are significant, and lack of fit is not significant against the pure error of the centre-point replicates, so they are the predictive basis of the design space. Step yield gave no significant model in either design, and it did not respond to any of the 4 factors over the ranges studied.

Every governed attribute meets its acceptance criterion everywhere in the characterized region. The worst case for viral clearance lies at the low edge of load pH and the high edge of load conductivity, with the other two parameters inside their ranges. There the fitted model predicts an MVM log reduction factor of 3.50 against a step requirement of 3.28 log10, a margin of about 0.7 residual standard deviations of the model. The normal operating range sits well inside that region, and the lowest MVM log reduction factor predicted anywhere inside it is 4.19.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter varying inside its normal operating range. The cumulative MVM clearance has a one-sided Cpk of 1.51, with a mean of 10.38 log10 against a requirement of 8.6 log10, which is the tightest capability of any A-Mab drug substance attribute. Pool host cell protein is far less constrained, with a Cpk of 6.1 against a limit of 100 ng/mg.

3 deviations were recorded, and each was investigated, impact assessed and closed (§13). One of them invalidated the first execution of both designs. The load material used in that execution had an out of trend acidic charge variant content, the designs were re-executed in full on a requalified load, and the analysis reported here is the re-executed one. The other two were resolved without change to the fitted effects, to the operating region or to the parameter classifications.

This step does not deliver the MVM requirement on its own. Small virus retentive filtration (PCR-009) contributes the larger single increment to the cumulative claim, and the two mechanisms are independent of each other. No claim is made here for the glycan or charge variant attributes, which are formed at the production bioreactor (PCR-003) and are not modified by anion exchange under platform conditions. The outcome of this report rolls up into the master report (PCMR-001).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced in a fed-batch mammalian cell culture and purified by a platform train of chromatographic and filtration steps. The train runs from Protein A capture through low-pH viral inactivation, cation exchange, anion exchange and small virus retentive filtration to ultrafiltration and diafiltration. Each step in the train has an assigned role in the control strategy, and the roles are set out in the master plan (PCMP-001). Anion exchange chromatography is Step 8 and the last chromatographic operation before the virus filter.

The step is operated in flow-through mode. The column is equilibrated and washed at low conductivity, the cation exchange eluate is titrated to the load pH and conductivity, and the antibody passes through the resin while acidic impurities bind, because under these conditions A-Mab carries a small net positive charge and does not bind. Host cell protein, DNA, leached Protein A and virus particles carry a net negative charge and are retained. The pool is collected across the load and the chase wash, and the column is then stripped and sanitized under its life-cycle procedure (SOP-2008).

Two things follow from that mechanism, and both shape this study. The separation depends on the difference in net charge between the product and the impurities, so it is governed by pH and by ionic strength and not by the elution conditions that govern a bind and elute step. The step also has no elution peak to cut, so the pool composition is set by how far collection runs into the trailing edge of the flow-through peak. The step is credited with clearance of minute virus of mice (a non-enveloped parvovirus), which is the hardest of the model viruses to remove. It is also credited with clearance of xenotropic murine leukaemia virus and with a further reduction of host cell protein, DNA and leached Protein A.

1.2 Objectives

The objectives of the characterization were as follows.

- (i) Quantify the relationship between the process parameters of the step and the quality attributes and performance attributes it governs, over ranges wider than the intended operating ranges.
- (ii) Define a multivariate operating region in which the governed attributes remain within their acceptance criteria.

- (iii) Establish proven acceptable ranges for each parameter against each attribute it governs, and confirm that the normal operating ranges sit inside them.
- (iv) Classify each process parameter according to the decision logic of SOP-4001, using the demonstrated effects and the capability with which the parameter can be controlled.
- (v) Provide the justification for the Stage 2 ranges and for this step’s contribution to the overall control strategy, for consolidation in the master report (PCMR-001).

1.3 Regulatory and scientific basis

The characterization is Stage 1 process design work in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), in which process understanding is built from prior knowledge and risk assessment and is then bounded by designed experiments. Criticality is treated as a continuum in the sense of ICH Q9 (International Council for Harmonisation 2023b), so an attribute is placed on a scale from very low to very high and the attributes of high and very high criticality are called critical. The same treatment is applied to parameters through the classification in §9.

The viral clearance work follows ICH Q5A(R2) (International Council for Harmonisation 2023a). Clearance is evaluated as a modular claim: each contributing step is studied independently on a qualified small-scale spiking model, and the increments are summed only where the mechanisms are independent. The A-Mab case study (CMC Biotech Working Group 2009) is the source of the process description, the quality attribute register and the risk methodology used here.

1.4 Related documents

The characterization package is cross-referenced. This report is the executed counterpart of PCP-008 and consumes the study scope set by RA-001.

Table 1.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-008	Process Characterization Plan (Anion Exchange Chromatography (Step 8))	Sibling — paired document

Docu- ment ID	Title	Relationship
PCMR- 001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The anion exchange polishing step is a platform operation at Novacyte Biologics, used in the manufacture of three previously licensed humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and in the supply of A-Mab clinical material. The resin, the buffer system and the mode of operation have not changed across those products. The accumulated experience shows that the step is robust in the sense that it delivers high yield and consistent impurity clearance when the load is presented at the platform pH and conductivity. That experience reduces the amount of optimization the A-Mab process required, and it sets the mechanistic expectations this study was designed to confirm or refute.

Four expectations were carried into the design. Raising the load pH increases the net negative charge on host cell protein and on the viral capsid, so it should increase both impurity binding and virus retention, and raising the conductivity of the load screens those same charges, so it should reduce clearance. The conductivity of the equilibration and wash-1 buffer sets the ionic environment in which weakly bound species are either retained or displaced during the chase, so it was expected to act mainly on pool host cell protein and only weakly on virus. Protein load determines how much of the resin capacity is used, so it was expected to matter only if the capacity for impurity binding were approached.

Prior knowledge also sets the boundaries of what this step is expected to do. Glycosylation variants are not separated by charge-based chromatography, except for sialylated structures, which are present in A-Mab at very low levels (CMC Biotech Working Group 2009). Charge variants are not modified by the step, since the operation is a flow-through polish at fixed pH and the product does not bind. Aggregate is reduced principally at the cation exchange step (PCR-007). None of these attributes was measured as a response here, and none is claimed for this step.

2.2 Quality attributes in scope

The step is assigned one critical quality attribute of its own, the clearance of minute virus of mice, which the register rates of very high criticality.

Table 2.1: Critical quality attribute assigned to the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20

The criticality in Table 2.1 is the Tool #1 score of the A-Mab framework, which combines the severity of the impact on the patient with the uncertainty in that assessment (CMC Biotech Working Group 2009). Attributes scored high or very high are treated as critical. Viral safety scores at the top of the scale because the consequence of a failure is severe and because clearance cannot be verified by testing the drug substance. The acceptance criterion in Table 2.1 is a cumulative requirement across the whole process, and the contribution required of this step is derived from it in §7.

The step also governs, or contributes to, four further attributes that are set or formed elsewhere in the process. They are listed below with the MVM row repeated from Table 2.1, so that every attribute the step acts on appears in one place.

Table 2.2: Quality attributes the anion exchange step controls or clears.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Leached Protein A	process impurity	0-5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20

Host cell protein and the two viral clearance attributes were measured as responses of the designed experiments, and residual DNA and leached Protein A were not. Both of the latter are cleared to well inside their limits before this step, principally at Protein A capture (PCR-005) and cation exchange (PCR-007), and both are measured in the anion exchange pool by validated methods (AMV-3014 and AMV-3016), with their drug substance capability reported in §8 alongside the rest. Making them responses of the design would have added two assays without adding process understanding, because neither is close to its limit at the entry to this step.

2.3 Risk-based prioritization and parameter selection

The scope of this study was set by the pre-characterization risk assessment (RA-001), which ranked every parameter of the step on its potential impact on a critical quality

attribute and on its potential to interact with other parameters. Parameters were then sorted into those warranting a multivariate study, those whose ranges could be supported by a univariate assessment, and those requiring no new work because prior knowledge already bounds them.

4 parameters were assigned to the multivariate design. Load pH and load conductivity were assigned because they set the charge difference the separation depends on, and because the risk assessment expected them to interact. Equilibration and wash-1 conductivity was assigned because it acts on the same binding equilibria at a different point in the cycle. Protein load was assigned because a capacity limit, if one exists inside the studied range, would show up as an interaction with conductivity and not as a main effect. Operating flow rate was assigned to a univariate assessment, because it acts on residence time and not on the ion exchange equilibria that the other four parameters set, and prior platform work has not linked it to a quality attribute at this step.

The characterization ranges are wider than the intended operating ranges by a factor of between 1.5 and 2.0. The widths were chosen so that the study bounds the process instead of confirming a point, and so that the knowledge space supports later movement within the design space. The upper edge of the load pH range was set below the point at which the product itself begins to bind, and the lower edge of the load conductivity range was set above the point at which the titrated cation exchange eluate becomes difficult to prepare reproducibly at scale. Both bounds are properties of the platform and not of A-Mab.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a laboratory-scale column qualified as a model of the commercial step under SOP-1001. The model uses the same resin, the same bed height and the same linear velocity as the commercial column, and the load, wash and chase volumes are normalized to column volumes. Protein load is expressed in grams of antibody per litre of resin, so it transfers between scales directly. Buffer compositions are identical to the commercial buffers, and the buffers used in the studies were released against the same specifications.

The qualification compared the scale-down model with commercial-scale data on the attributes that enter and leave the step. Pool host cell protein, residual DNA, leached Protein A, step yield and pool conductivity were compared at the target condition, and the column was compared on plate count and peak asymmetry before each study. The model is judged qualified because no difference between scales was found on those attributes that exceeded the reproducibility of the methods. The qualification record is held with the study documentation for PCP-008 and is summarized in the master plan (PCMP-001).

The viral clearance work used the same column operated as a small-scale spiking model in the sense of ICH Q5A(R2) (International Council for Harmonisation 2023a). The load was spiked with XMuLV or MVM to a titre high enough that the log reduction factor can be measured against the assay limit of quantitation, and the spiked material was clarified before loading so that the spike did not change the pressure behaviour of the column. Spiking runs are performed in a containment laboratory and cannot be run at commercial scale, so the clearance claim rests on the qualification of the model rather than on at-scale confirmation. This is the principal limitation of the viral clearance claim and it is stated again in §11.

The centre-point replicates of both designs are the pure error term used to test lack of fit in §5, which is the second purpose of the qualification. Their spread is reported in §5.1 and it sets the resolution of every effect estimate in this report.

3.2 Operation

The step was operated as it is operated at commercial scale, under SOP-2011. The column is sanitized and equilibrated with the equilibration buffer, whose conductivity is one of the studied parameters. The cation exchange eluate is titrated to the load pH and adjusted to the load conductivity, filtered and loaded at the operating flow

rate. The load is followed by a chase wash with the same buffer used for equilibration, which is the reason the buffer is described here as the equilibration and wash-1 buffer. The flow-through pool is collected from the start of the product breakthrough on the ascending edge of the UV trace until the trailing edge falls to the collection set-point. The column is then stripped, sanitized and stored under SOP-2008.

Several inputs to the step are controlled to platform specifications and were not varied. The resin lot, the buffer compositions, the load filter, the column dimensions and the sanitization regime are identity and quality controlled, not characterized, and the temperature of the operation is held at ambient room conditions at both scales. The one input attribute that proved to matter is the charge variant content of the cation exchange eluate, which is a property of the upstream step and not a parameter of this one. It is discussed in §13.1 and it is the reason a feed input control is proposed in §10.

3.3 Analytical methods

Every response was measured by a validated method. The controlled documents that govern the step and its assays are listed in Table 3.1.

Table 3.1: Controlled documents governing the anion exchange characterization.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuvLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

Pool host cell protein was measured by the process-specific ELISA (AMV-3012) and is reported in nanograms per milligram of antibody. XMuLV and MVM titres were measured by infectivity assay (AMV-3017 and AMV-3018), and the log reduction factor is the difference between the log₁₀ virus load applied to the column and the log₁₀ virus recovered in the pool, corrected for volume. Step yield was determined from the antibody mass in the pool relative to the mass loaded, by ultraviolet absorbance. Residual DNA and leached Protein A were measured in the pool by AMV-3014 and AMV-3016 respectively. The charge variant content of the load material was measured by imaged capillary isoelectric focusing (AMV-3013), which is the method that detected the deviation described in §13.1. Validated performance for these methods is summarized in Appendix D.

3.4 Sampling plan

Each design run produced a load sample, a flow-through pool sample and a strip sample. The load sample establishes the input attributes and, for the spiking runs, the input virus titre. The pool sample carries the responses. The strip sample was retained but was tested only where a mass balance was needed to close an investigation. Every run in both designs was executed as a spiking run, so a log reduction factor is reported for each of the 19 screening conditions and each of the 28 response-surface conditions.

The screening design carries 3 centre-point replicates and the response-surface design carries 4. They were distributed through the run order and not grouped, so that a drift in the assays or in the feed would show up as variation between replicates, and run order within each design was randomized under SOP-1002.

3.5 Statistical methods

The factors were coded so that the midpoint of the characterization range is zero and the two edges are minus one and plus one, so effects are expressed per half-range, which makes factors with different units comparable. The screening data were fitted by ordinary least squares with all main effects and all two-factor interactions, and an effect is reported as twice the coded coefficient, which is the change in the response across the full studied range of that factor. The response-surface data were fitted with the full quadratic model in the same 4 factors, including the two-factor interactions and the squared terms.

A term is called significant at a significance level of 0.05, and model adequacy is judged on four quantities together: the coefficient of determination, the adjusted value that penalises added terms, the predicted value computed from the PRESS statistic, and the overall model F test. Lack of fit was tested by partitioning the residual sum of squares into a lack of fit component and a pure error component estimated from the centre-point replicates, and a model whose lack of fit is not significant is treated as an adequate description of the surface. A factor with no significant term in either

design is called robust over the studied range, and that conclusion is reported and not dropped.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every process parameter drawn inside its normal operating range. Capability is reported as a one-sided Cpk against the acceptance criterion that applies to the attribute. No numeric acceptance criterion for Cpk itself is defined for this campaign, so the capability figures in §8 are read together with the margin between the simulated mean and the limit.

4 Study design

4.1 Factors, ranges and the knowledge space

The parameters carried into the study, their set-points, their normal operating ranges and their characterization ranges are given in Table 4.1. The characterization range is the knowledge space edge, which is the widest condition at which the step was run. It is not a proven acceptable range, and the two are distinguished throughout this report.

Table 4.1: Process parameters, ranges, study type and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	univariate

The classification column in Table 4.1 is the outcome of §9 and is shown here so that the table is complete. The four multivariate factors are coded A to D throughout the report and the appendices, using the legend in Table 4.2.

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

The centre point of both designs is the midpoint of each factor’s characterization range, and it was replicated so that the reproducibility of the step and of the assays

could be estimated independently of the model. Those replicates are the pure error term used in the lack of fit tests.

4.2 Screening design

The screening study was a two-level full factorial in the 4 factors with 3 centre points, giving 19 runs. A full factorial was chosen over a fractional design because the risk assessment (RA-001) expected interactions between pH and the two conductivities, and a resolution IV fraction would have aliased those interactions with each other. The design estimates every main effect and every two-factor interaction without confounding.

The design is nonetheless near-saturated, spending most of its degrees of freedom on the model and leaving few for residual error, so the residual is estimated from few degrees of freedom and the adequacy statistics that depend on it are unstable (§5.2). The purpose of the screening study is to identify which factors act on which response, and that is how its results are used in §5.2. It is not the predictive model. The full coded design matrix and the measured responses are given in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same 4 factors, with 4 centre points and 28 runs. The axial points sit on the faces of the cube rather than outside it, so the design does not require the step to be run beyond the characterization ranges. Face centring costs some rotatability and it is the right trade here, because the characterization ranges were chosen as the limits of what the step should be asked to do.

All 4 screening factors were carried into the response-surface design. Reducing the factor set is the usual practice after a screening study, but two of the four factors act on host cell protein and a different pair acts on the two viral responses, so dropping either pair would have removed a governing factor for one of the governed attributes. Protein load appears in neither pair, and it was retained so that the capacity question could be bounded over the whole load range. Its absence of effect in the response-surface design is the evidence that the resin capacity is not approached (§5.4). The design adds the curvature terms, it is not near-saturated, and it supports interpolation inside the region. It is the predictive model and the basis of the design space in §6, and its coded matrix and responses are in Appendix B.

4.4 Univariate assessment

Operating flow rate was assessed one factor at a time across its characterization range with the other parameters at their set-points. It was excluded from the multivariate

design for a mechanistic reason. Flow rate changes the residence time of the load on the column, and the binding of host cell protein and of virus to a strong anion exchanger under these conditions is fast relative to the residence time at any flow rate in the studied range. It does not change the charge difference that the separation depends on, so it cannot interact with pH or conductivity in the way those three parameters interact with each other.

No effect estimate for flow rate on any governed attribute is reported in this document, and the univariate runs are held in the study record for PCP-008. The classification of the parameter in §9 therefore rests on its mechanism and on the control the skid applies to it under SOP-2011, and not on a measured effect.

5 Results

5.1 Centre-point performance and reproducibility

The step reproduces well at its centre point, and the least reproducible response is pool host cell protein. Table 5.1 gives the mean, standard deviation and coefficient of variation of the 4 centre-point replicates of the response-surface design.

Table 5.1: Centre-point reproducibility of the response-surface design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	13.7	2.48	18.2
XMULV LRF ($\log_{10}$)	4	6.63	0.445	6.71
MVM LRF ($\log_{10}$)	4	4.9	0.336	6.86
Step yield	4	0.966	0.0123	1.27

Pool host cell protein varies by 18.2 percent between replicates run at identical conditions, the two viral responses by 6.7 and 6.9 percent, and step yield by 1.3 percent. A large part of the host cell protein variation is analytical: the ELISA (AMV-3012) has an intermediate precision of 9.5 percent relative standard deviation, and the method register attributes a fraction of 0.4 of the observed variance in this attribute to the method. The residual variation is the step itself, which is consistent with a flow-through operation whose pool composition depends on where collection stops.

The two infectivity assays are more precise in relative terms than the ELISA because they report a logarithm. The MVM assay (AMV-3018) has an intermediate precision of 0.32 $\log_{10}$ and the XMULV assay (AMV-3017) 0.3 $\log_{10}$. The centre-point standard deviation for MVM is 0.34 $\log_{10}$, which is of the same order as the assay itself, so it bounds the step's own variation in MVM clearance and does not measure it. The screening design gives the same picture at its own centre point, where host cell protein varies by 20.6 percent and the MVM standard deviation is 0.27 $\log_{10}$ (Appendix A). Reproducibility is of the same order in both designs.

5.2 Screening: factor effects

The screening study resolved which factors act on which response, and it separated the two governing pairs cleanly. It did not produce a usable model for every response, and the model summaries are given in Table 5.2.

Table 5.2: Screening model summaries for the four measured responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.966	0.923	0.841	22.6	8.55e-05	2.45
XMULV LRF (log ₁₀)	19	0.842	0.644	-0.106	4.25	0.0257	0.393
MVM LRF (log ₁₀)	19	0.97	0.932	0.869	25.8	5.18e-05	0.178
Step yield	19	0.498	-0.13	-3.64	0.792	0.642	0.0126

3 of the 4 responses gave a significant model, and step yield did not: its adjusted coefficient of determination is negative, which means the fitted terms explain less than the mean alone. The predicted coefficient of determination for XMULV clearance is also negative, so that fit describes the runs it was built from and should not be used to predict a new condition. Both figures are what a near-saturated two-level design produces when the effects are small relative to the residual error, and neither is a reason to distrust the effect estimates for the responses whose models are significant. The screening design identifies the active factors. The response-surface model in §5.3 is the predictive model and the basis of the design space.

Pool host cell protein is governed by the equilibration and wash-1 conductivity and by the load pH, and the two interact.

Table 5.3: Largest screening effects on pool host cell protein.

Term	Effect	p-value	signif.
wash1_cond	12.1	9.48e-06	yes
load_ph	-11.9	1.05e-05	yes
load_ph × wash1_cond	-6.63	0.000642	yes
load_cond	-1.66	0.213	
load_ph × load_cond	1.65	0.214	
wash1_cond × load_cond	-1.21	0.353	

Raising the wash-1 conductivity across its studied range raised pool host cell protein by 12.1 ng/mg ($p < 0.001$). Raising the load pH across its studied range lowered it by 11.9 ng/mg ($p < 0.001$). The interaction between them is -6.6 ng/mg ($p < 0.001$), and its sign means that the benefit of a higher load pH is larger when the wash-1 conductivity is at the top of its range. Neither load conductivity nor protein load produced a resolved effect on pool host cell protein. The absence of a protein load effect is the more informative of the two, because it says that the capacity of the resin for host cell protein is not approached anywhere in the studied range.

Both viral responses are governed by the load pH and the load conductivity, and by neither of the other two factors.

Table 5.4: Largest screening effects on the MVM log reduction factor.

Term	Effect	p-value	signif.
load_ph	1.22	7.83e-07	yes
load_cond	-0.715	4.32e-05	yes
wash1_cond	0.182	0.0757	
wash1_cond $\times$ load	0.107	0.263	
load_ph $\times$ load	-0.0627	0.502	
load	-0.0451	0.626	

Table 5.5: Largest screening effects on the XMuLV log reduction factor.

Term	Effect	p-value	signif.
load_ph	1	0.000934	yes
load_cond	-0.605	0.0151	yes
load	0.268	0.209	
load_cond $\times$ load	-0.255	0.23	
load_ph $\times$ load	0.25	0.24	
wash1_cond	0.198	0.342	

Raising the load pH across its studied range raised the MVM log reduction factor by 1.22 log10 ($p < 0.001$), and raising the load conductivity lowered it by 0.71 log10 ($p < 0.001$). The corresponding figures for XMuLV are 1.00 and 0.60 log10, and no interaction reached significance for either virus in the screening design. The largest term after the two main effects is the equilibration and wash-1 conductivity on MVM, at 0.18 log10 with $p = 0.076$, which is not significant at the stated level and is small against the centre-point standard deviation of 0.34 log10.

Step yield produced no significant effect from any term, its largest estimate being for load conductivity at -0.012 ($p = 0.102$) against a centre-point mean of 0.966. Yield is therefore robust to all 4 factors over the ranges studied. All 4 are retained in the knowledge space, and the full effect table is in Appendix C.

5.3 Response-surface models

The response-surface models describe the three governed responses adequately and are the predictive basis of the design space. Table 5.6 gives the summaries.

Table 5.6: Response-surface model summaries for the four measured responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.934	0.864	0.66	13.2	1.83e-05	2.65

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
XMuLV LRF (log ₁₀)	28	0.915	0.824	0.621	10	8.69e-05	0.314
MVM LRF (log ₁₀)	28	0.898	0.787	0.394	8.14	0.000269	0.32
Step yield	28	0.488	-0.0632	-2.18	0.885	0.589	0.0145

For the three governed responses the coefficient of determination lies between 0.898 and 0.934, the adjusted value does not fall below 0.787, and every model F test is significant. The predicted coefficient of determination is lowest for MVM at 0.394. That statistic measures how well the model predicts a run it has not seen. It is the reason the design space claim in §6 is stated at the level of the mean response and is not extended to an assurance statement at the edge of the region. Step yield again gave no significant model, consistent with the screening result.

Lack of fit is not significant for any of the three governed responses, the smallest lack of fit p-value being 0.499, and the partitions for the MVM and host cell protein models are given below.

Table 5.7: Analysis of variance for the MVM response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	11.7	14	0.834	8.14	0.000269
Residual	1.33	13	0.102	nan	nan
Lack of fit	0.994	10	0.0994	0.881	0.619
Pure error	0.339	3	0.113	nan	nan

Table 5.8: Analysis of variance for the pool host cell protein response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1,302.7	14	93.1	13.2	1.83e-05
Residual	91.5	13	7.04	nan	nan
Lack of fit	73.0	10	7.3	1.19	0.499
Pure error	18.5	3	6.16	nan	nan

In both cases the lack of fit mean square is of the same order as the pure error mean square. The quadratic surface therefore describes the data no worse than replicate runs describe each other, which is the condition under which a response surface may be used to interpolate. The coefficient tables follow.

Table 5.9: Full quadratic coefficients for the MVM log reduction factor.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	4.85	0.111	43.9	1.63e-15	***
A	0.539	0.0754	7.15	7.48e-06	***
B	0.0307	0.0754	0.407	0.691	
C	-0.564	0.0754	-7.48	4.65e-06	***
D	0.0261	0.0754	0.346	0.735	
AB	0.127	0.08	1.59	0.136	
AC	-0.074	0.08	-0.925	0.372	
AD	0.0353	0.08	0.441	0.666	
BC	-0.0339	0.08	-0.424	0.679	
BD	0.00805	0.08	0.101	0.921	
CD	-0.0156	0.08	-0.195	0.848	
A ²	-0.322	0.199	-1.61	0.13	
B ²	0.149	0.199	0.747	0.469	
C ²	0.0224	0.199	0.112	0.912	
D ²	0.135	0.199	0.679	0.509	

The MVM surface is close to planar. Load pH contributes 0.539 log10 per half-range and load conductivity -0.564 log10 per half-range, both with $p < 0.001$ and $p < 0.001$ respectively. No interaction and no squared term reaches significance, and the largest curvature term is the square of load pH at -0.322 ($p = 0.130$), which indicates a slight flattening of the clearance gain at the top of the pH range but does not support a claim of an optimum inside the range.

Table 5.10: Full quadratic coefficients for pool host cell protein.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	14.8	0.917	16.2	5.51e-10	***
A	-5.34	0.625	-8.54	1.09e-06	***
B	6.19	0.625	9.89	2.04e-07	***
C	0.0399	0.625	0.0637	0.95	
D	0.387	0.625	0.619	0.546	
AB	-1.45	0.663	-2.19	0.0475	*
AC	-0.925	0.663	-1.39	0.187	
AD	0.0192	0.663	0.029	0.977	
BC	0.15	0.663	0.226	0.825	
BD	-0.559	0.663	-0.843	0.415	
CD	-0.124	0.663	-0.187	0.855	
A ²	1.91	1.65	1.16	0.268	
B ²	2.22	1.65	1.34	0.202	
C ²	-0.0187	1.65	-0.0113	0.991	
D ²	-1.98	1.65	-1.2	0.253	

The host cell protein surface is steeper and it is not planar. The wash-1 conductivity

coefficient is 6.19 ng/mg per half-range and the load pH coefficient is -5.34 ng/mg per half-range, both with $p < 0.001$. Their interaction survives into the response-surface model at -1.45 ng/mg per half-range ($p = 0.047$). The screening design put the same term at -3.32 ng/mg per half-range, so the response-surface model confirms the interaction in sign at a smaller estimated magnitude. The two curvature terms with the largest estimates are the squares of load pH and wash-1 conductivity, and neither is significant. The coefficients for the XMuLV model and for step yield are in Appendix C.

The surfaces themselves are shown over the two factor pairs that matter, beginning with Figure 5.1, which gives the four responses over load pH and load conductivity, the plane the viral responses live in.

Response-surface predictions (remaining factors held at set-point)

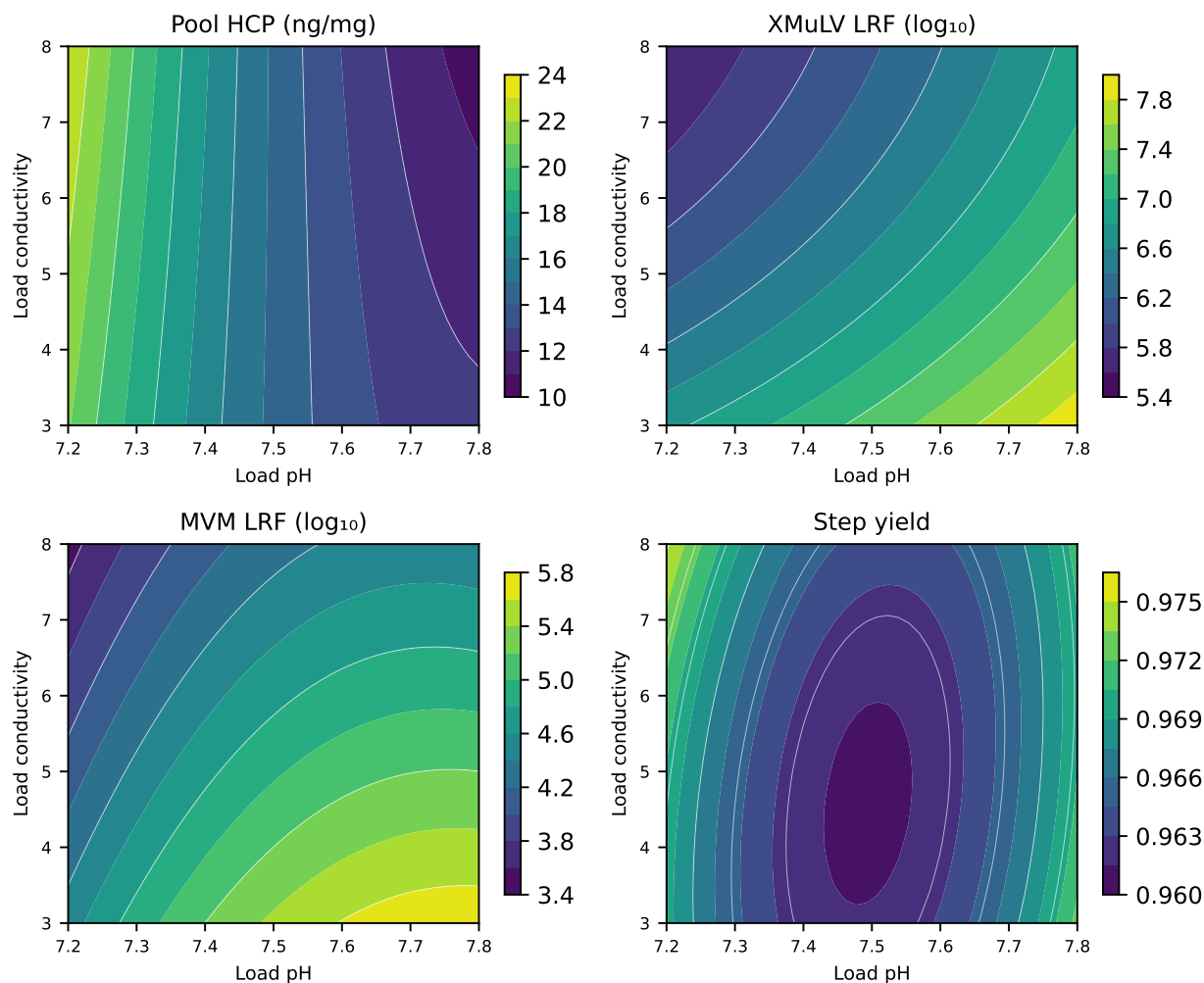


Figure 5.1: Response-surface predictions over load pH and load conductivity, with the remaining two factors held at set-point.

The viral panels show parallel diagonal contours, which is what a model with two main effects and no interaction looks like. Clearance is highest at high load pH and low load conductivity and falls smoothly towards the opposite corner. The host cell protein panel in the same plane is almost flat, because neither of these two factors governs it.

The plane that governs host cell protein is load pH against the equilibration and wash-1 conductivity, and it is shown in Figure 5.2.

Response-surface predictions (remaining factors held at set-point)

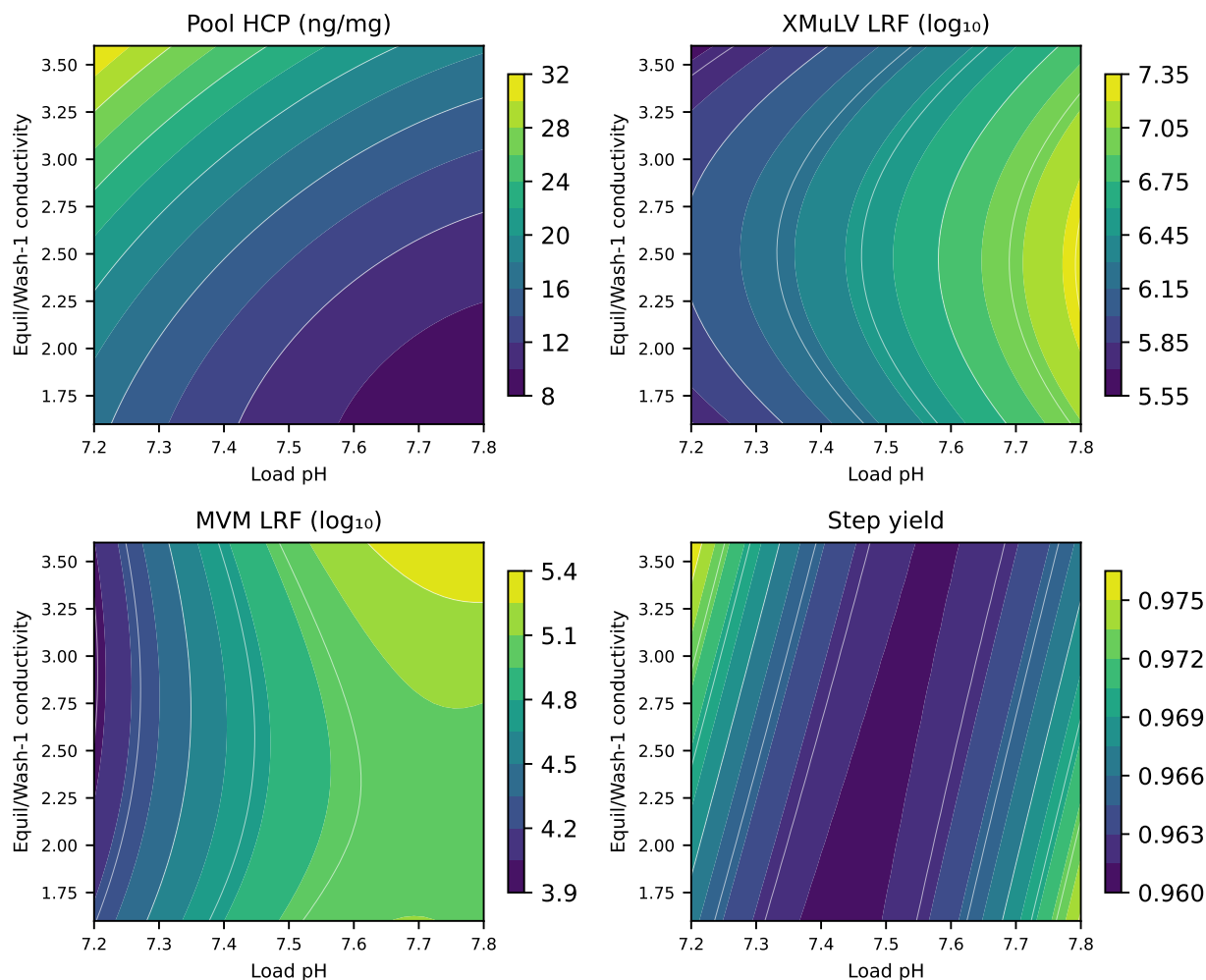


Figure 5.2: Response-surface predictions over load pH and equilibration and wash-1 conductivity, with the remaining two factors held at set-point.

Here the host cell protein contours converge towards the low conductivity edge, which is the graphical form of the interaction in Table 5.10. The viral panels in this plane are nearly flat, and the two pictures together are the reason the operating region is described in §6 by two planes and not by one.

Model diagnostics for the MVM response are given in Figure 5.3. The residuals show no structure against the predicted value, the normal plot is close to linear over the body of the distribution, and the actual against predicted plot lies on the identity line with no systematic departure at either end. Diagnostics for the other responses were reviewed to the same standard and showed the same behaviour.

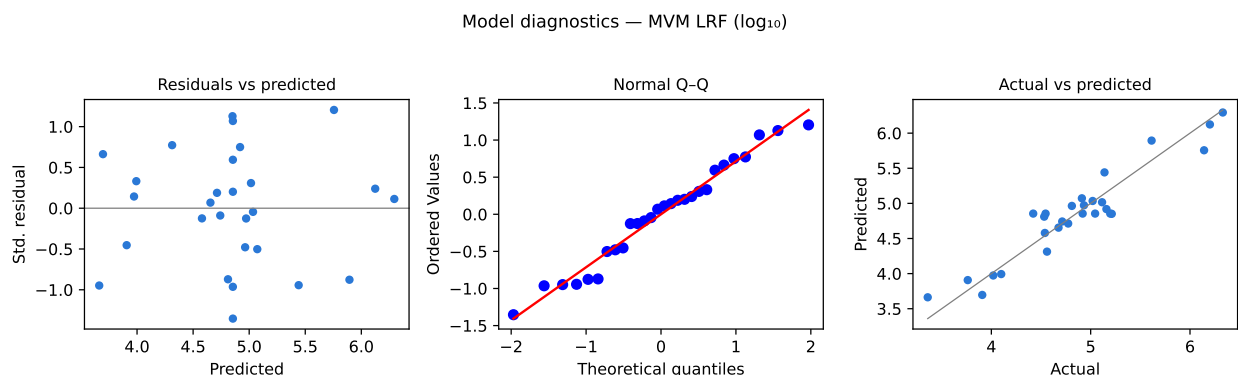


Figure 5.3: Model diagnostics for the MVM response-surface model: residuals against predicted, normal quantile plot, and actual against predicted.

5.4 Mechanistic interpretation

The results follow the mechanism the step was expected to work by, and the two governing pairs separate for a physical reason. Virus clearance depends on the binding of the capsid to the resin during the load, so it responds to the pH and the ionic strength of the load and to nothing else in the studied set. Host cell protein clearance depends on the same binding equilibria, but the species involved are a heterogeneous population with a wide range of isoelectric points, and the weakly bound part of that population is displaced during the chase wash. That is why host cell protein responds to the conductivity of the equilibration and wash-1 buffer while virus does not.

The interaction between load pH and wash-1 conductivity has the same explanation. At low wash conductivity the weakly bound host cell protein species stay on the column whatever the load pH was, so the pH effect is muted. At high wash conductivity those species are displaced into the pool, and the load pH then decides how strongly they were bound in the first place. The consequence for control is direct: the two parameters must be considered together, and a proven acceptable range for either one that ignores the other would misstate the risk.

The absence of a protein load effect on any response is the most useful negative result of the study. If the resin capacity for impurity binding were being approached at the top of the load range, the clearance responses would fall and the effect would appear as an interaction between load and conductivity. No such term is significant in either design. The step therefore has capacity margin across the whole studied load range, and the load can be used as an operational lever without a quality consequence. The

one caveat is that the studied range is bounded, and this study says nothing about loads above its upper edge.

Step yield is high and flat because the product does not bind. In a flow-through operation the only routes to product loss are non-specific interaction with the resin and incomplete recovery of the pool, and neither is sensitive to the parameters studied here over these ranges. The predicted yield at the set-point is 0.961, which is consistent with the centre-point mean of 0.966.

6 Design space

The design space for this step is the whole characterized region in the 4 multivariate parameters. Every governed attribute is predicted to meet its acceptance criterion everywhere in that region, so the region is not cut back by any attribute inside the ranges studied. This is a stronger result than the study was designed to produce.

The region has two principal planes. Viral clearance is governed by load pH and load conductivity, and host cell protein by load pH and the equilibration and wash-1 conductivity (§5.4). Load pH appears in both, which makes it the parameter that ties the region together. Protein load appears in neither, so it enters the region only as the range over which the other three were studied.

The worst case for the region is not one of its corners. The MVM model carries positive squared terms in the equilibration and wash-1 conductivity and in the protein load (Table 5.9), so its lowest predicted value lies inside the region and not at a vertex, and the region was searched on a refined grid. Neither squared term is significant on its own, and the search was made over the fitted model as it stands. The worst point sits at load pH 7.2, equilibration and wash-1 conductivity 3.04 mS/cm, load conductivity 8 mS/cm and protein load 185 g/L resin. There the model predicts an MVM log reduction factor of 3.50 against the step requirement of 3.28 log₁₀ derived in §7, and an XMuLV log reduction factor of 5.00 against 3.78 log₁₀. The MVM margin there is 0.22 log₁₀, which is about 0.7 times the residual standard deviation of the fitted model (0.32 log₁₀). The worst case for host cell protein is at a different point, where the model predicts 33.0 ng/mg against a limit of 100 ng/mg.

The normal operating ranges sit inside the characterized region with room to spare. Searching the normal operating box the same way, the lowest predicted MVM log reduction factor is 4.19, a margin of 0.91 log₁₀ or about 2.8 residual standard deviations. The lowest predicted XMuLV log reduction factor in the same box is 5.83 and the highest predicted pool host cell protein is 22.3 ng/mg. The nesting is the conventional one: the normal operating range lies inside the design space, which lies inside the knowledge space bounded by the characterization ranges (CMC Biotech Working Group 2009).

Three bounds apply to this claim and none of them is removable by more analysis of these data. The models predict mean levels, so at the edge of the region the reliability of an individual batch is lower than the mean statement suggests, and the margin at the worst case is smaller than the model's own residual standard deviation. The claim covers the four parameters that were varied, at the ranges in Table 4.1, with the load presented at the charge variant content of a representative cation exchange eluate. It rests on a qualified scale-down model and, for the viral responses, on a small-scale spiking model that cannot be run at commercial scale (§3.1).

Movement within the design space is not a regulatory change in the sense of ICH Q8 (International Council for Harmonisation 2009), although it remains subject to the pharmaceutical quality system and to a prospective assessment of the particular move (International Council for Harmonisation 2008). Movement outside it requires change control and, for a parameter that governs viral clearance, a new assessment of the modular claim.

7 Proven acceptable ranges

The design space in §6 is a multivariate region, and it is complemented here by proven acceptable ranges, which are per attribute and per parameter, because a control strategy has to be written in terms a batch record can carry.

The acceptance basis differs by attribute. For pool host cell protein the criterion is the drug substance specification, an upper limit of 100 ng/mg. For the two viral attributes the drug substance criterion is a cumulative log reduction across the process, so the criterion that applies to this step is the required step contribution, obtained by subtracting the clearance credited to the other steps from the cumulative requirement. For MVM the cumulative requirement is 8.6 log₁₀, the other credited steps contribute 5.32 log₁₀, and the requirement on this step is therefore 3.28 log₁₀. For XMuLV the same calculation gives 3.78 log₁₀, and these step level criteria are used throughout this section and in §6.

Two analyses were run for each attribute and parameter pair. The first holds the other parameters at their set-points and evaluates the fitted response-surface model across the parameter's characterization range. The second propagates the normal operating ranges of the other parameters: they are drawn by Monte-Carlo inside their normal operating ranges, the fitted model is evaluated with the model's residual variation added, and the parameter is acceptable where the whole 95 percent predictive interval stays within the criterion. The second analysis is a robustness criterion, so it is the narrower of the two by construction. Both are reported in Table 7.1.

Table 7.1: Proven acceptable ranges per governed attribute and parameter, from both analyses.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
Pool HCP (ng/mg)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
Pool HCP (ng/mg)	Load conductivity	3–8	mS/cm	3–8	3–8
Pool HCP (ng/mg)	Protein load	50–300	g/L resin	50–300	50–300
XMuLV LRF (log ₁₀)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
XMuLV LRF (log ₁₀)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
XMuLV LRF (log ₁₀)	Load conductivity	3–8	mS/cm	3–8	3–8
XMuLV LRF (log ₁₀)	Protein load	50–300	g/L resin	50–300	50–300
MVM LRF (log ₁₀)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
MVM LRF (log ₁₀)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
MVM LRF (log ₁₀)	Load conductivity	3–8	mS/cm	3–8	3–8
MVM LRF (log ₁₀)	Protein load	50–300	g/L resin	50–300	50–300

For every attribute and every parameter the two analyses agree, and both span the full characterization range. No parameter has a sub-range within which the governed attribute fails, and no parameter loses acceptable range when the other three are allowed to vary within their normal operating ranges. The MVM rows are the ones to read first, because MVM carries the tightest requirement of the three attributes, and even there the acceptable range for load pH runs from the bottom to the top of the studied range.

That outcome has a consequence for how the ranges should be used. Since no univariate proven acceptable range is binding, the constraint on the operating region is the multivariate worst case identified in §6 and not any single parameter limit in Table 7.1. A batch record that held each parameter inside its row of Table 7.1 independently would not be relying on a proven constraint, because the rows were computed with the other parameters at their set-points or inside their normal operating ranges, not at their own extremes.

The representative plots follow, one per governed attribute, each against the parameter with the largest main effect on it. In each figure the acceptance limit is drawn, the acceptable region is shaded green, and the set-point and normal operating range are marked, with the left panel giving the first analysis and the right panel the second.

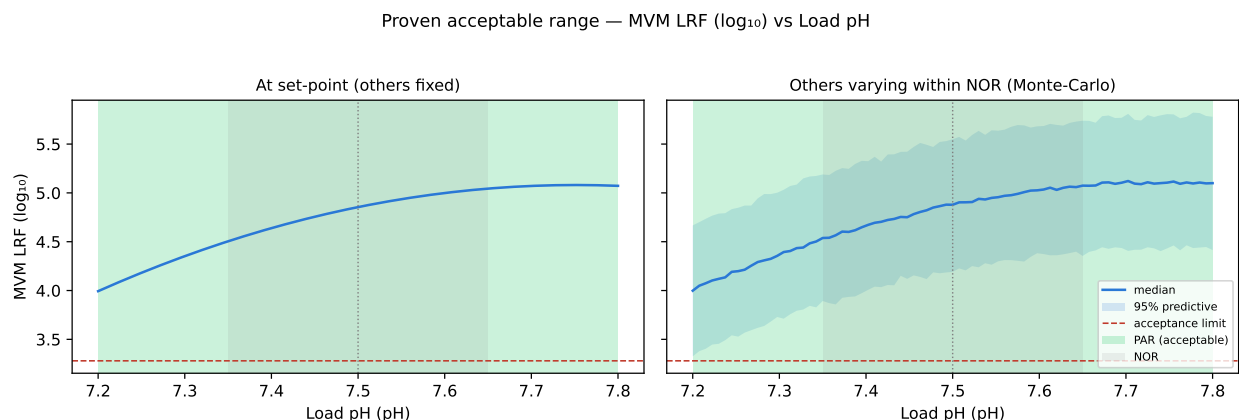


Figure 7.1: Proven acceptable range for the MVM log reduction factor against load pH, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

Figure 7.1 shows the MVM log reduction factor rising with load pH across the whole studied range and staying above the step requirement throughout, in both analyses. In the right panel the lower bound of the predictive interval stays above the step requirement across the whole pH range, with the least margin at the low edge.

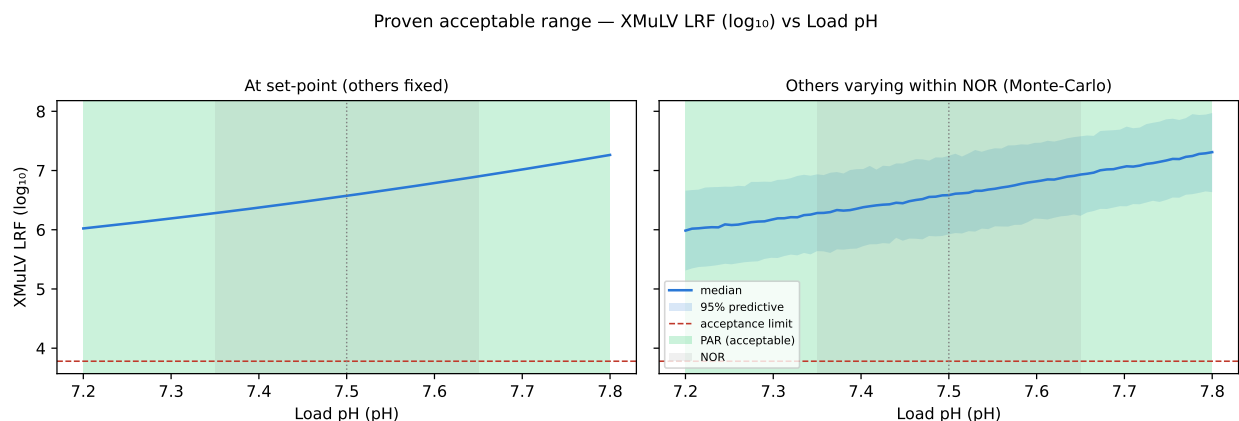


Figure 7.2: Proven acceptable range for the XMuLV log reduction factor against load pH, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

Figure 7.2 shows the same shape with a wider margin, because XMuLV is an enveloped virus cleared by three steps in the train, so the requirement placed on this step is a smaller part of what the step achieves.

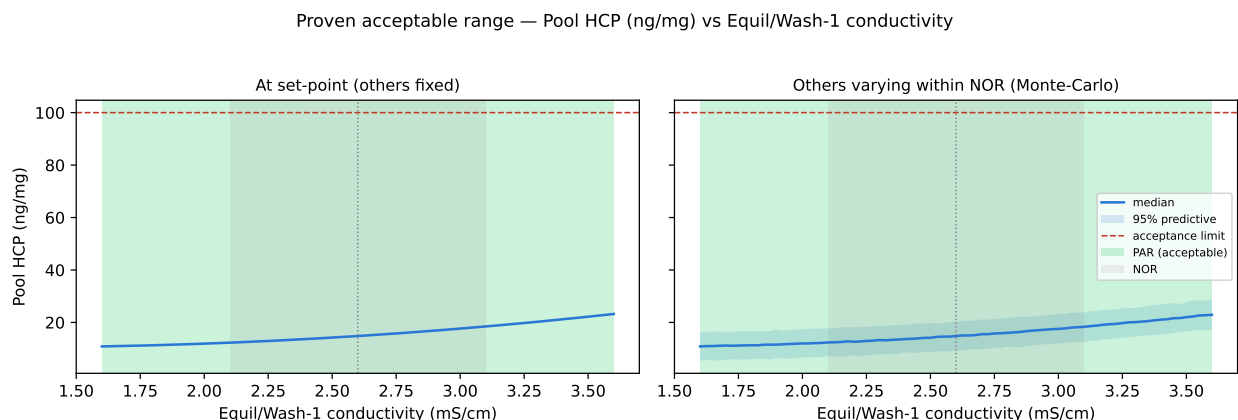


Figure 7.3: Proven acceptable range for pool host cell protein against the equilibration and wash-1 conductivity, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

Figure 7.3 is the clearest picture of how much margin the step carries on host cell protein. The response rises with wash-1 conductivity, as the effect estimates predict, but the whole studied range sits far below the limit and the upper bound of the predictive interval does not approach it. Host cell protein does not constrain the operating region of this step.

These ranges are bounded by the characterization ranges and by the fitted models, so they are not evidence about conditions outside Table 4.1, and they carry the assumptions of §6 on the scale-down model and on the charge variant content of the load.

8 Process capability and robustness

Commercial-scale capability was estimated by simulating 2,000 batches with every process parameter in the train drawn inside its normal operating range, by the method in §3.5, and the attributes this step governs or contributes to are given in Table 8.1.

Table 8.1: Commercial-scale capability for the attributes the anion exchange step governs.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 $\pm$ 4.6	6.1
Residual DNA	VL	upper	0-0.001	0.0001 $\pm$ 0	10.8
Leached Protein A	L-M	upper	0-5	0.003 $\pm$ 0.0006	2,785.3
Viral clearance — XMuLV (enveloped)	VH	lower	16.7-30	19.5 $\pm$ 0.61	1.5
Viral clearance — MVM (parvovirus)	VH	lower	8.6-20	10.4 $\pm$ 0.4	1.5

The cumulative MVM clearance is the tightest capability in Table 8.1 and the tightest of any A-Mab drug substance attribute, with a Cpk of 1.51. Its simulated mean is 10.38 log10 with a standard deviation of 0.40, against a requirement of 8.6 log10, and the lowest simulated batch was 9.08 log10. The cumulative XMuLV clearance is next at a Cpk of 1.53 with a mean of 19.51 log10. The margin is the more informative number of the two, because no numeric Cpk acceptance criterion is defined for this campaign (§3.5).

Host cell protein carries a Cpk of 6.1, with a simulated mean of 15.2 ng/mg and a worst simulated batch of 42.3 ng/mg against a limit of 100 ng/mg. Residual DNA and leached Protein A are further from their limits still, at 10.8 and 2785. Those two attributes are cleared principally upstream of this step, and the capability shown for them is a property of the train and not of the anion exchange operation.

The tightest capability of the drug substance is therefore an attribute this step contributes to but does not deliver alone. Of the cumulative 10.03 log10 of MVM clearance, this step is credited with 4.71 log10 and small virus retentive filtration with 5.32 log10 (PCR-009). Low-pH inactivation is credited with 0 log10 against MVM, because a non-enveloped parvovirus is not inactivated by low pH (PCR-006). The clearance credited to this step agrees with the scale-down model: the response-surface model predicts 4.87 log10 at the set-point, and the difference of 0.16 log10 between the two is smaller than the standard deviation of the centre-point replicates.

Capability is bounded in three ways. It is estimated with parameters inside their normal operating ranges and not at the edges of the design space. It uses the fitted models from a qualified scale-down model, and for the viral attributes from a small-scale spiking model. It is a prediction to be confirmed at commercial scale during process performance qualification, which has not been executed.

9 Parameter classification

Every parameter of the step is classified well controlled critical process parameter, and no parameter is classified critical process parameter. The distinction follows SOP-4001 and the A-Mab decision logic: both classes cover parameters whose variability can affect a critical quality attribute, and the difference is the risk of leaving the design space (CMC Biotech Working Group 2009). That risk is low for every parameter here, because the design space extends to the full characterized range and the normal operating ranges in Table 4.1 sit inside it.

- Load pH is a well controlled critical process parameter with the largest main effect on both viral responses and the second largest on host cell protein, and it appears in both governing planes of the design space. It is classified well controlled rather than critical because the titration is performed in a controlled vessel with an in-line measurement, and because the acceptable range extends across the whole characterized range and does not stop inside it.
- Equilibration and wash-1 conductivity is a well controlled critical process parameter. It has the largest main effect on pool host cell protein and it interacts with load pH. It is prepared by buffer dilution and is verified on release, so an out of range value is detected before use, which is what happened in the deviation described in §13.3.
- Load conductivity is a well controlled critical process parameter. It has the second largest main effect on both viral responses and no resolved effect on host cell protein. It is set by titration of the cation exchange eluate and is measured in line before loading.
- Protein load is a well controlled critical process parameter, although no effect of protein load on any governed attribute was resolved in either design, which is evidence that the resin capacity is not approached within the studied range. The classification is retained rather than reduced because the parameter sets how much impurity is presented to a finite number of binding sites, so its effect is bounded by the studied range and not by mechanism. It is easy to control, since it is computed from the measured mass of the load.
- Operating flow rate is a well controlled critical process parameter. It was assessed univariately, and this report presents no effect estimate for it (§4.4). The classification rests on the mechanism, in that flow rate sets residence time, and on the control the skid applies to the parameter under SOP-2011, over a normal operating range that lies inside the range characterized (Table 4.1).

No parameter of this step is classified key process parameter or general process parameter, which is a consequence of the step's role and not of the size of the effects.

Every parameter of a step that carries a viral clearance claim is linked to a critical quality attribute through the clearance mechanism, even where the measured effect is small, and the counts across the whole train are consolidated in PCMR-001.

10 Contribution to the control strategy

The step contributes four elements to the control strategy, and it delivers none of them in isolation.

The first is parameter control. The 5 parameters are held within the normal operating ranges in Table 4.1, which sit inside the design space established in §6. The two that carry the viral clearance claim, load pH and load conductivity, are set by titration of the cation exchange eluate and are verified before the load begins.

The second is in-process monitoring. Pool host cell protein is measured on every batch by AMV-3012 and trended, and residual DNA and leached Protein A are measured by AMV-3014 and AMV-3016. Pool conductivity, pool pH and step yield are recorded as process attributes. The UV trace of the flow-through peak is recorded, and the pool collection set-point on the trailing edge is a fixed value in the batch record following the correction described in §13.2.

The third is the viral clearance claim itself, which is modular: this step is credited with 4.71 log₁₀ of MVM clearance and 5.95 log₁₀ of XMuLV clearance. The claim is additive only because the mechanisms are independent: charge-based partitioning here, size-based retention at the virus filter (PCR-009), and chemical inactivation at low pH (PCR-006). Against MVM the low-pH step contributes nothing, so the cumulative claim of 10.03 log₁₀ rests on this step and the virus filter together. The consolidated claim is reported in PCMR-001.

The fourth is a feed input control that arises from a deviation, since the charge variant content of the cation exchange eluate is an input attribute of this step, and an extended neutralized hold of that eluate raised it enough to change the behaviour of the step (§13.1). A hold time limit on the cation exchange eluate before anion exchange loading is therefore proposed as part of the control strategy, and the icIEF method (AMV-3013) is available to verify the input where a hold excursion occurs.

Life-cycle controls sit alongside these, in that resin re-use is limited and the column is repacked and re-qualified under SOP-2008, with a qualification that includes the plate count and asymmetry criteria used in this study. Resin life-cycle performance was not a factor of these designs, and the clearance claim is made for a column within its validated re-use limit.

The step does not modify the glycan or charge variant attributes of the product, which are formed at the bioreactor (PCR-003). It is not the principal aggregate polish, which is the cation exchange step (PCR-007), and it does not by itself meet the cumulative viral clearance requirement for either model virus.

11 Discussion

The step is well understood at the end of this study, and the understanding is mechanistic rather than empirical. Two parameters govern virus clearance, a partly different pair governs host cell protein, and the reason for the split follows from the chemistry of a flow-through anion exchanger (§5.4). The main effects are the effects the prior knowledge section expected, and they act in the directions it expected. The interaction is the one expectation the study did not meet. The risk assessment expected load pH and load conductivity to interact, and no such term was resolved in either design, while the interaction that governs the step is between load pH and the equilibration and wash-1 conductivity. The full factorial was still the right design, because it could have resolved either.

The scale-down model is the foundation of every quantitative claim in this report, and its representativeness is supported in two ways. The qualification compared input and output attributes between scales and found no difference beyond method reproducibility (§3.1). The centre-point replicates then reproduced within a few percent for every response except host cell protein, where a fraction of 0.4 of the variance is attributed to the ELISA. Neither of those is a demonstration that the model predicts the commercial column at the edges of the design space, and that limitation is general to Stage 1 work (CMC Biotech Working Group 2009).

The first of the three deviations is genuinely adverse. The first execution of both designs used a load whose acidic charge variant content was 33.5 percent against 21 percent in the requalified material, and at the high load and high wash conductivity corner it drove pool host cell protein above the acceptance criterion applied to it. The designs were invalidated for operating region definition and re-executed in full, the root cause is understood, the anomalous interaction is absent from the requalified data, and the reported analysis uses only the requalified execution (§13.1). The residual position is that this step is sensitive to an input attribute that was not previously controlled, and the control strategy now carries a hold time limit for that reason.

Three limitations bound what the report claims. The viral clearance results come from a small-scale spiking model that cannot be run at commercial scale, so the clearance claim is a modular claim supported by a qualified model and not by at-scale measurement. The response-surface models predict mean levels, and their predicted coefficients of determination, the lowest being 0.394, set the limit of what should be read from a prediction at an untested condition. Resin life-cycle was not a variable of the study, so the claims hold for a column within its validated re-use limit and the life-cycle evidence sits in SOP-2008.

No parameter of this step required the most stringent control class. The reason is not that the effects are small, because two of them are large, but that the acceptable region extends to the edges of the characterization ranges and the parameters are

held far inside it. A critical process parameter designation would be appropriate if a design space edge fell inside the range the equipment can reach, and no such edge was found here.

12 Conclusions

The anion exchange step is characterized and its operating region is defined. The 4 multivariate parameters were studied over ranges between 1.6 and 2.0 times the width of their normal operating ranges, and response-surface models for pool host cell protein and for both viral responses are adequate and free of significant lack of fit.

The design space is the full characterized region. Every governed attribute meets its acceptance criterion everywhere in it, and the worst case is the MVM log reduction factor at 3.50 log₁₀ against a step requirement of 3.28 log₁₀. The proven acceptable ranges span the full characterization range for every attribute and parameter pair, under both analyses, so the binding constraint on the operating region is the multivariate worst case and not a univariate limit.

At commercial scale the attributes this step governs meet their acceptance criteria, and the tightest capability is the cumulative MVM clearance at a Cpk of 1.51 with a mean of 10.38 log₁₀ against 8.6 log₁₀. All 5 parameters are classified well controlled critical process parameters and none is a critical process parameter.

3 deviations were recorded and closed. One invalidated the first execution of both designs, which were re-executed in full, and none of the three changed a parameter classification or the operating region. The clearance credited to this step is one increment of a modular claim that also rests on small virus retentive filtration and, for enveloped virus, on low-pH inactivation. Confirmation at commercial scale is a Stage 2 activity and has not been performed, and these results roll up into the master report (PCMR-001).

13 Deviations from the plan

3 deviations from PCP-008 were recorded during execution, and each was investigated, its root cause determined, its impact on the study assessed and its disposition approved. One of them invalidated the first execution of both designs and is described first.

Table 13.1: Deviation register for the anion exchange characterization.

De- via- tion	Summary	Detected during	Disposition
DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	invalidated_and_re_executed
DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected_by_modelling_and_verification_runs
DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	retained

Table 13.1 lists the three. The first has a disposition of invalidation and re-execution, which is the most severe of the three dispositions and is treated at the greatest length below. The second was corrected by modelling and verification runs and applies to both executions, and the third was retained, which means the run was kept in the analysis with a documented justification.

13.1 DEV-008-01: non-representative load in the first execution

The first execution of both designs produced pool host cell protein above the acceptance criterion applied to it. The highest value in the first screening execution was 150 ng/mg against the criterion of 100 ng/mg, and the highest in the first response-surface execution was 130 ng/mg. 3 of the 47 runs across the two first executions exceeded the criterion, and all of them lay at or near the corner combining high protein load with high equilibration and wash-1 conductivity. In the re-executed designs reported here the highest pool host cell protein at any condition is 35.7 ng/mg.

The investigation began from the load material, and charge variant analysis by icIEF (AMV-3013) of the load used in the first execution, lot LOT-CEX-8842, returned an acidic variant content of 33.5 percent against 21 percent for the requalified load. The cation exchange eluate had been held after neutralization for 36 hours before titration, and the root cause was assigned as asparagine deamidation promoted by that extended hold. Deamidation lowers the isoelectric point of the affected species, which increases their affinity for an anion exchanger at the load pH and puts them in competition with the host cell protein population for the same binding sites.

The mechanism predicts a specific statistical signature, and the data carry it. In the first execution protein load had a large effect on pool host cell protein and it interacted strongly with the equilibration and wash-1 conductivity, and neither term survives in the requalified data.

Table 13.2: Screening effects on pool host cell protein in the first (invalidated) and requalified executions.

Term	Effect (first)	p (first)	Effect (requalified)	p (requalified)
B	50.4	0.000195	12.1	9.48e-06
D	35.6	0.00181	0.885	0.491
BD	34.8	0.00211	0.224	0.86
A	-22.7	0.0194	-11.9	1.05e-05

Table 13.2 compares the four largest terms of the first execution with the same terms in the requalified execution. The protein load effect falls from 35.6 ng/mg ($p = 0.002$) to 0.9 ng/mg ($p = 0.491$), and the interaction between protein load and wash-1 conductivity falls from 34.8 ng/mg ($p = 0.002$) to 0.2 ng/mg ($p = 0.860$). The wash-1 conductivity effect, which is a property of the step rather than of the load, is present in both executions and changes only in magnitude. Competition for a finite number of binding sites is exactly what produces a load by conductivity interaction, and its disappearance when the competing species are removed confirms the root cause.

The disposition was to invalidate both designs for the purpose of defining the operating region and to re-execute them in full on a requalified, representative load. The re-executed datasets are the ones analysed in §5 to §9 of this report, and the first execution is retained as a superseded dataset, referenced here for root cause confirmation and not used for any quantitative claim about the step. Its value is that it characterizes the behaviour of the step when the load is not representative, which is information the control strategy uses (§10).

Two consequences follow. A hold time limit on the neutralized cation exchange eluate before anion exchange loading is proposed as a feed input control. Where a hold excursion occurs, the charge variant content of the load is to be verified by AMV-3013 before the step is run, because the first execution shows that an out of trend load changes which parameters matter and not only how well the step performs.

13.2 DEV-008-02: pool collection set-point on the trailing edge

The pool collection set-point used on the trailing edge of the flow-through peak was more permissive than the value the method now specifies. Both executions collected to 180 mAU, and the set-point was subsequently corrected to 120 mAU. The deviation was found during the post-execution reconciliation of pool fractions against the host cell protein results (AMV-3012), after both designs had been executed.

The impact is a common offset and not a distortion of the design. The same set-point was applied to every run of both executions, so it shifts the intercept of every fitted model and leaves the estimated factor effects unchanged. Pool host cell protein is the response affected, because it is the response that depends on how far collection runs into the trailing edge. The viral responses are computed from the ratio of virus in the load to virus in the pool, and both quantities move with the pool volume, so the log reduction factor is far less sensitive to the collection window.

The correction was evaluated by applying the fitted models at the corrected collection window and confirmed with 3 verification runs, of which 2 were placed at the extremes of the operating region and not at the centre. The verification set is small. With 3 runs and the intermediate precision of the host cell protein ELISA, the 95 percent confidence half-width on the verification mean is 23.6 percent of that mean. The verification therefore confirms that the corrected set-point produces host cell protein of the same order as the models predict, and it does not resolve a difference smaller than that half-width. That resolution is sufficient here only because the margin to the limit is large, at a simulated mean of 15.2 ng/mg against 100 ng/mg (§8). The disposition is that the corrected set-point is fixed in the batch record and the reported operating region is unchanged.

13.3 DEV-008-03: equilibration and wash-1 buffer released below its pH range

One screening run was executed with an equilibration and wash-1 buffer that had been released outside its pH acceptance range. Buffer lot LOT-BUF-8815 measured 7.32 against a lower acceptance limit of 7.4 and a target of 7.5. The deviation was found during buffer release testing under SOP-2103, and the root cause was a titration endpoint error at buffer preparation.

The run was retained in the analysis. The buffer pH is not a factor of the design, but it sets the pH of the column at the start of the load, so the excursion acts in the same direction as a lower load pH. The measured value lies inside the load pH characterization range of 7.2 to 7.8, near its lower edge, over which the effect of pH on every response was quantified. The argument for retaining the run rests on the direction of the bias. Lower pH reduces both viral clearance and host cell protein clearance, which means the excursion biases the affected run towards worse apparent performance and cannot flatter the result.

The effect on the fitted models is one run out of 19, at a shift smaller than the half-range of the load pH factor, on a response whose centre-point coefficient of variation is 20.6 percent. The disposition is retained without adjustment, a corrective action was raised against the buffer preparation procedure, and the release test that detected the excursion is unchanged and remains the detection control.

14 Appendices A to D

The appendices carry the full design matrices, the complete effect and coefficient tables, and the analytical method summary. Nothing in them is new, and they are the primary data behind §5.

14.1 Appendix A: screening design matrix and responses

The coded settings of the 19 screening runs are in Table 14.1 and the measured responses in Table 14.2. The factor letters are those of Table 4.2.

Table 14.1: Screening design, coded factor settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 14.2: Screening design, measured responses.

Run	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
1	12.5	6.16	4.59	0.99

Run	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
2	15.6	6.15	4.43	0.975
3	12.1	5.47	3.76	0.951
4	14.6	5.56	3.96	0.964
5	35.7	6.34	4.74	0.965
6	35	6.3	4.83	0.995
7	29.2	5.92	4.01	0.974
8	29.8	5.95	3.94	0.969
9	9.62	7	5.91	0.973
10	7.37	7.4	5.61	0.953
11	8.69	6.61	5.25	0.972
12	8.04	6.71	4.89	0.973
13	12.7	6.59	5.91	0.995
14	15	8.34	5.97	0.981
15	12.9	6.69	5.15	0.966
16	15	6.52	5.31	0.965
17	16.2	6.42	4.54	0.966
18	17.5	6.8	5.08	0.975
19	11.6	5.95	4.83	0.974

14.2 Appendix B: response-surface design matrix and responses

The coded settings of the 28 response-surface runs are in Table 14.3 and the measured responses in Table 14.4.

Table 14.3: Response-surface design, coded factor settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1

Run	Type	A	B	C	D
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 14.4: Response-surface design, measured responses.

Run	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
1	13.5	6.72	4.55	0.951
2	17.3	6.03	5.21	0.989
3	14.1	5.87	4.02	0.995
4	15	5.3	3.76	0.951
5	28.5	6.53	4.77	0.988
6	25.5	6.71	4.71	0.963
7	31.3	5.36	3.91	0.995
8	32.3	5.06	3.36	0.966
9	5.66	7.5	6.14	0.974
10	8.37	7.83	5.61	0.989
11	7.56	6.49	4.54	0.972
12	7.43	7.13	4.68	0.988
13	17.6	7.18	6.2	0.974
14	18	8.12	6.33	0.968
15	13.9	6.58	4.53	0.96
16	13.9	6.37	5.16	0.973
17	23	5.83	4.1	0.971
18	12	7.37	4.91	0.965
19	8.1	6.42	4.93	0.946
20	27.5	6	5.02	0.972
21	15.6	7.44	5.14	0.952
22	15.5	5.93	4.56	0.967
23	13	6.52	4.81	0.968
24	14.3	6.7	5.11	0.952
25	15	6.59	4.42	0.958
26	10.7	6.03	5.2	0.975
27	16.3	7.05	5.04	0.977

Run	Pool HCP (ng/mg)	XMuLV LRF (log ₁₀)	MVM LRF (log ₁₀)	Step yield
28	12.6	6.86	4.92	0.953

14.3 Appendix C: full effect and coefficient tables

The complete screening effect tables for all 4 responses follow. Effects are per full studied range of the factor, and the significance column marks the conventional thresholds.

Table 14.5: Screening effects, pool host cell protein (all terms).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	
CD	0.261	0.13	0.613	0.212	0.837	
BD	0.224	0.112	0.613	0.182	0.86	

Table 14.6: Screening effects, XMuLV log reduction factor (all terms).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	
BD	0.126	0.0632	0.0982	0.643	0.538	
AC	-0.0968	-0.0484	0.0982	-0.493	0.635	
AB	-0.094	-0.047	0.0982	-0.478	0.645	
BC	-0.0169	-0.00844	0.0982	-0.0859	0.934	

Table 14.7: Screening effects, MVM log reduction factor (all terms).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	
AC	0.0135	0.00677	0.0446	0.152	0.883	
AB	-0.0132	-0.00659	0.0446	-0.148	0.886	

Table 14.8: Screening effects, step yield (all terms).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.0116	-0.00582	0.00315	-1.85	0.102	
B	0.00743	0.00371	0.00315	1.18	0.272	
AD	-0.00715	-0.00357	0.00315	-1.13	0.29	
AC	0.00513	0.00257	0.00315	0.815	0.439	
BC	-0.00392	-0.00196	0.00315	-0.622	0.551	
BD	0.00383	0.00192	0.00315	0.608	0.56	
CD	0.00346	0.00173	0.00315	0.549	0.598	
AB	0.00141	0.000707	0.00315	0.224	0.828	
D	-0.00136	-0.000679	0.00315	-0.215	0.835	
A	-0.000663	-0.000332	0.00315	-0.105	0.919	

The response-surface coefficients for MVM and for pool host cell protein are given in §5.3. The remaining two responses follow.

Table 14.9: Response-surface coefficients, XMuLV log reduction factor.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.57	0.109	60.6	2.51e-17	***
A	0.621	0.074	8.39	1.33e-06	***
B	-0.076	0.074	-1.03	0.323	
C	-0.554	0.074	-7.48	4.6e-06	***
D	0.0287	0.074	0.388	0.704	
AB	-0.0278	0.0785	-0.355	0.729	
AC	0.0223	0.0785	0.285	0.78	
AD	0.192	0.0785	2.44	0.0295	*
BC	-0.118	0.0785	-1.5	0.158	
BD	0.0577	0.0785	0.735	0.475	
CD	-0.0751	0.0785	-0.957	0.356	
A ²	0.0696	0.195	0.356	0.727	
B ²	-0.328	0.195	-1.68	0.118	

Term	Coef.	Std. err.	t	p-value	Sig.
C ²	0.152	0.195	0.779	0.45	
D ²	0.0761	0.195	0.39	0.703	

Table 14.10: Response-surface coefficients, step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.961	0.00501	192	7.92e-24	***
A	-0.000294	0.00342	-0.086	0.933	
B	0.00022	0.00342	0.0643	0.95	
C	0.00109	0.00342	0.318	0.756	
D	-0.00203	0.00342	-0.594	0.563	
AB	-0.00464	0.00362	-1.28	0.223	
AC	-0.00175	0.00362	-0.482	0.637	
AD	0.00608	0.00362	1.68	0.117	
BC	-0.000152	0.00362	-0.042	0.967	
BD	-0.00433	0.00362	-1.2	0.253	
CD	-0.00413	0.00362	-1.14	0.275	
A ²	0.00932	0.00902	1.03	0.32	
B ²	0.000524	0.00902	0.0581	0.955	
C ²	0.00152	0.00902	0.168	0.869	
D ²	0.00177	0.00902	0.196	0.847	

14.4 Appendix D: analytical methods summary

The validated performance of the methods used at this step is summarized in Table 14.11. Precision is intermediate precision, reported as a relative standard deviation for the quantitative assays and as a log10 standard deviation for the two infectivity assays.

Table 14.11: Validated performance of the analytical methods used at the anion exchange step.

Method Title		Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3016	Leached Protein A by ELISA	6.5 %RSD	0.25 ppm	97.4	0.41
AMV-3017	XMULV Infectivity Titre (TCID50)	0.3 log10 SD	1.5 log10 TCID50/mL	96	0.28
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32 log10 SD	1.4 log10 TCID50/mL	96	0.3
AMV-3012	Host-Cell Protein by ELISA	9.5 %RSD	1 ng/mg	95	0.4

The methods listed in Table 3.1 but absent from Table 14.11 (AMV-3014, AMV-3013) do not carry a performance record in the campaign method register, and their validation reports are the reference for their performance.

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